IGI LIMS

An illustrated tour of the specimen tracking system, as built to date

PIRL, UNC · October 13, 2026 · status: prototype, specification in flight

1. What it is

IGI LIMS is a chain-of-custody database and web interface for the biological specimens collected in the IGI cancer clinical trial. It records one row per tube. Every tube stores the tube it was made from, so the path from a collected specimen, through fractionation, cryopreservation, extraction and library preparation, to the sequencing or mass-spectrometry data file is a chain of parent links that can be walked in either direction. It runs as a single Flask application on SQLite with no external services, sized for a lab of about two dozen users on a workstation or a GCP virtual machine.

The screenshots in this document were taken from a throwaway demonstration database. The participant, tubes, users and lists shown are invented. Red numbered arrows mark the features explained under each picture.

2. The tube identifier

Identifiers are composed by the database from the tube's attributes and are never typed by hand. The format is defined in one configuration list so that it can change as the specification settles without touching the schema.

```
IGI-CC-PPPPPPP-YYMMDD-SS-FFF-MMM-AA (35 characters)
```

Segment	Meaning	Validation
IGI	Project constant	literal
CC	Cohort	active code in the cohort list (MP, AM, TN, NC)
PPPPPPP	Participant code	seven upper-case letters, no I or O; generated
YYMMDD	Collection date	parsed as a calendar date
SS	Specimen type	active code in the specimen list (PF, PB, BM, TU, LN, ...)
FFF	Fraction	active code in the fraction list (BLK, PLA, PBM, PFM, CD8, ...)
MMM	Material	active code in the material list (NAT, CPU, DNA, HMD, LIB, PEP, ...)
AA	Aliquot number	01 to 99, issued per prefix by the database, never reused

The worked example that follows produces `IGI-MP-NDAKVTX-261006-PF-BLK-NAT-01`: cohort MP, participant NDAKVTX, collected 6 October 2026, pleural fluid, unfractionated, native, first aliquot. Its children keep the first five segments and change only fraction, material and aliquot.

3. What the system enforces

- **Controlled vocabularies.** Every code used in IDs, forms and metadata lives in one code-list table managed on the Codes page. Codes are retired rather than deleted, so historical records keep their meaning.
- **No free text except Notes.** Operator, SOP, storage location, custody parties and purposes, and every metadata vocabulary (kits, lots, buffers, media, antibodies, instruments) are dropdowns. The server rejects

values that are not active list entries. The only remaining free-text fields are machine identifiers (data file URI, run ID, checksum) and the participant's external site ID.

- **Derivation rules.** A rule table decides which fractions a specimen type may have, which fraction and material transitions are allowed, and which materials a root tube may arrive as. A violation blocks creation and offers an override request instead.
- **Two-person agreement.** Overrides, and by default every edit, delete or restore of a tube record, go to an approval queue. The requester cannot approve their own request. The approver's decision applies the change.
- **Soft deletes and a full audit log.** Deleting a tube keeps the record and its aliquot number, and is refused while the tube has active children. Every write is recorded with who, when, and before and after values.

4. Walkthrough: signing in and finding your way

4.1 Sign in

Every page requires a login. Accounts are created by an admin from the Users page or the command line.

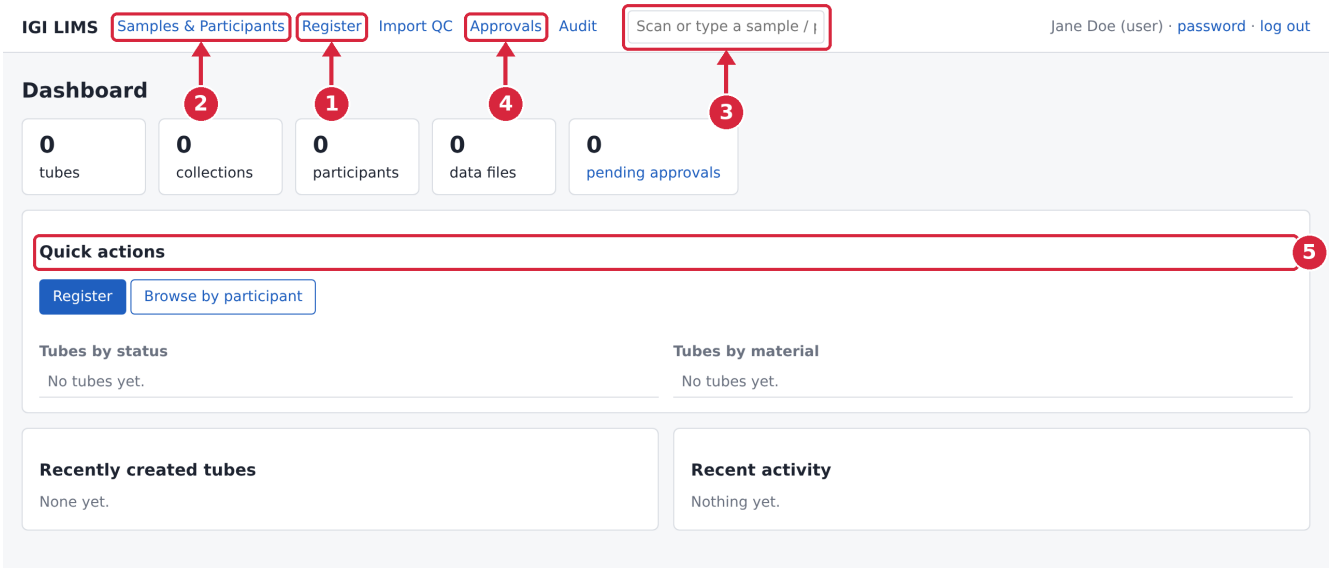


1 Username (the account name, not the display name)

2 Sign in

4.2 The dashboard

The dashboard after signing in. Nothing on it is editable; it links to the three working areas.

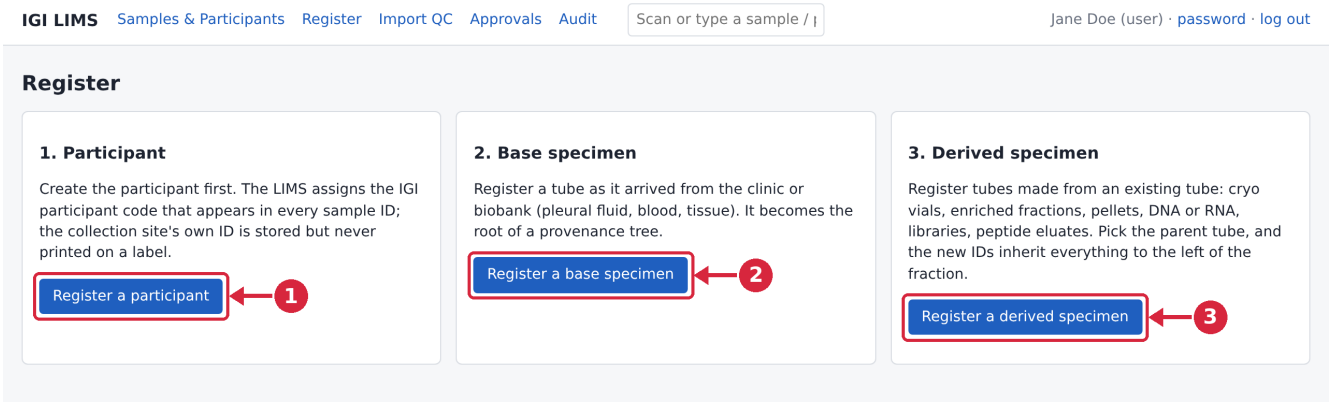


1. Register: the only place records are created
2. Samples & Participants: read-only browsing
3. Jump box: scan or paste a tube ID to open it
4. Approvals queue; the badge counts pending requests
5. Quick actions, counts, recent tubes and recent activity

4.3 The Register page

All record creation starts here. The three cards are the three kinds of record a lab member creates, and the order matches the order of events: a participant enrols, a specimen is collected, tubes are derived.

The Register page offers exactly three actions, in the order they happen in the lab.

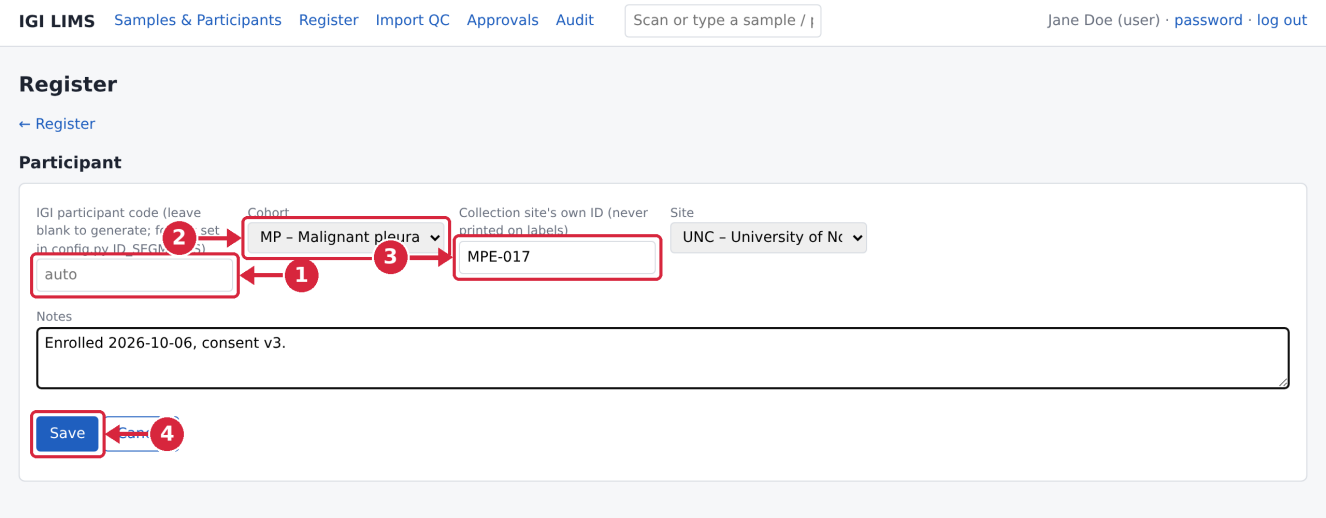


1. Step 1: a person enrolled in the trial
2. Step 2: a tube as collected (root of a tree)
3. Step 3: a tube made from an existing tube

5. Walkthrough: registering a participant and a base specimen

5.1 Register a participant

Registering a participant. Only the cohort is mandatory; the participant code is issued by the system.



1 Left blank: the seven-letter code is generated (no I or O)

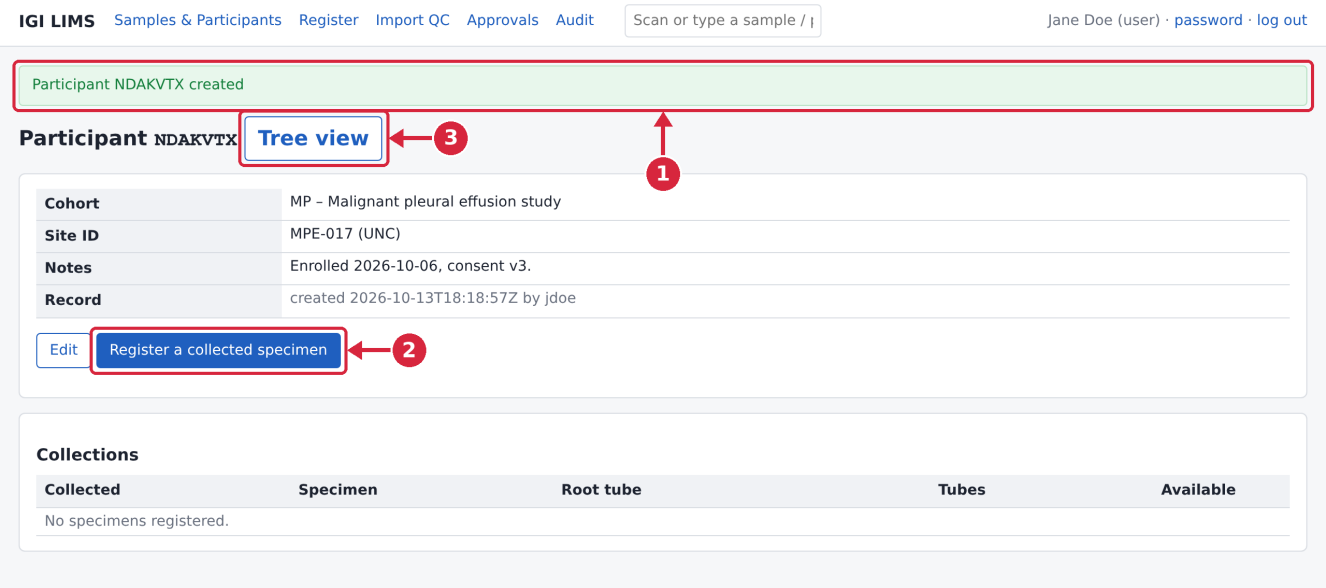
2 Cohort, from the cohort code list

3 The site's own ID for cross-reference

4 Create

5.2 The participant record

The participant record. It lists every collection and links to the tree.



1 Confirmation with the generated code

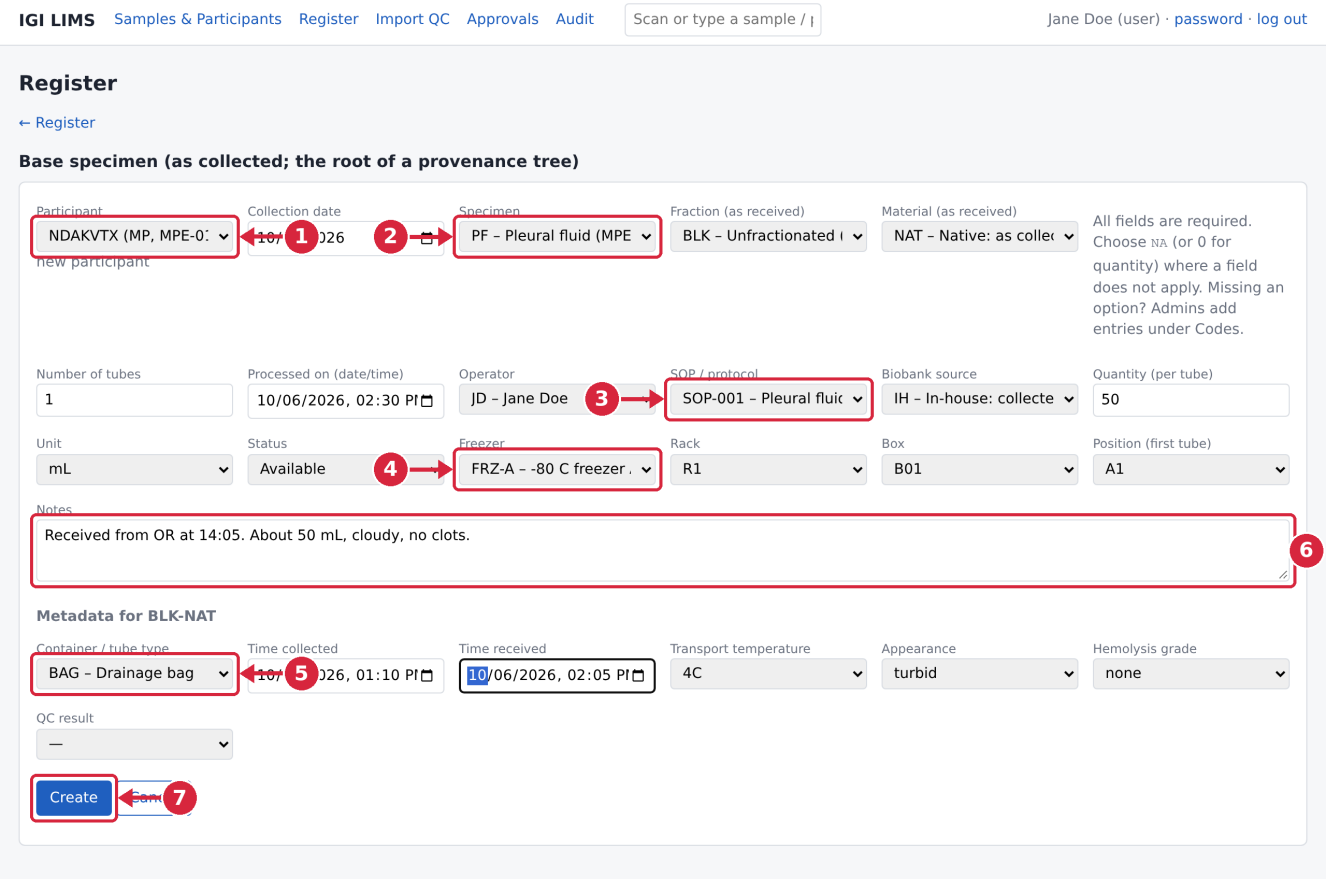
2 Go straight to registering this participant's first tube

3 Opens the read-only tree for this participant

5.3 Register the collected specimen (the root tube)

A base specimen is a tube with no parent. Its specimen type, fraction and material as received become segments of the ID. The metadata block at the bottom changes with the fraction and material chosen; for an unfractionated fluid it asks for container, collection and receipt times, transport temperature, appearance and hemolysis.

Registering the collected pleural fluid. All fields are required; NA is an explicit choice where a field does not apply.



1 Participant chosen on the previous page

2 Specimen, fraction and material as received: these become ID segments

3 Every field is a dropdown; lists are maintained under Codes

4 Storage location: freezer, rack, box, position

5 Metadata fields appear for the chosen fraction and material

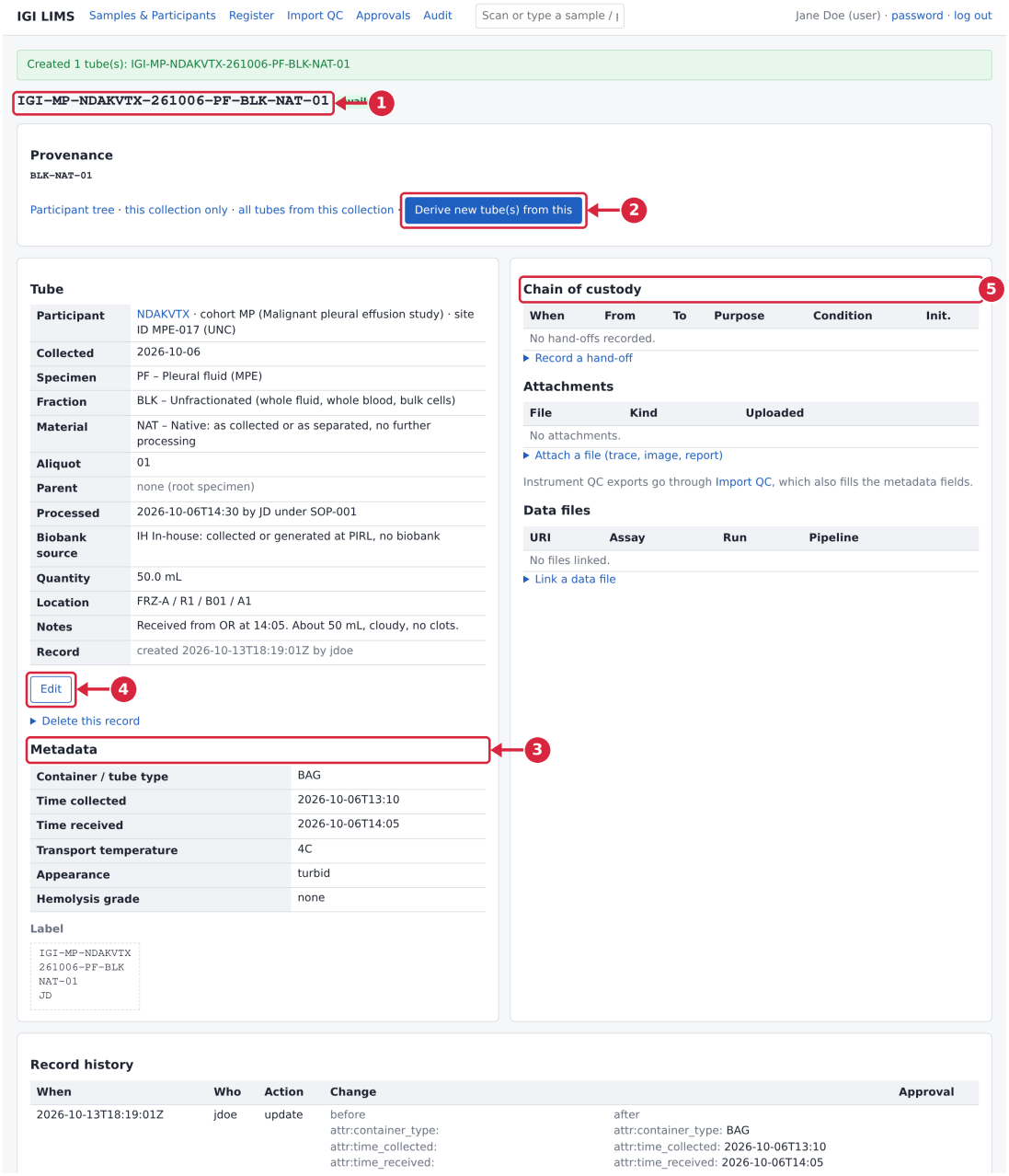
6 Notes: the only free-text field in the system

7 Create: the tube ID is composed by the database

5.4 The tube page

Every tube has one page: identity and provenance at the top, then the tube's own fields, metadata, a printable label layout, custody events, attachments, linked data files, and its full change history.

The tube page for the new root specimen.



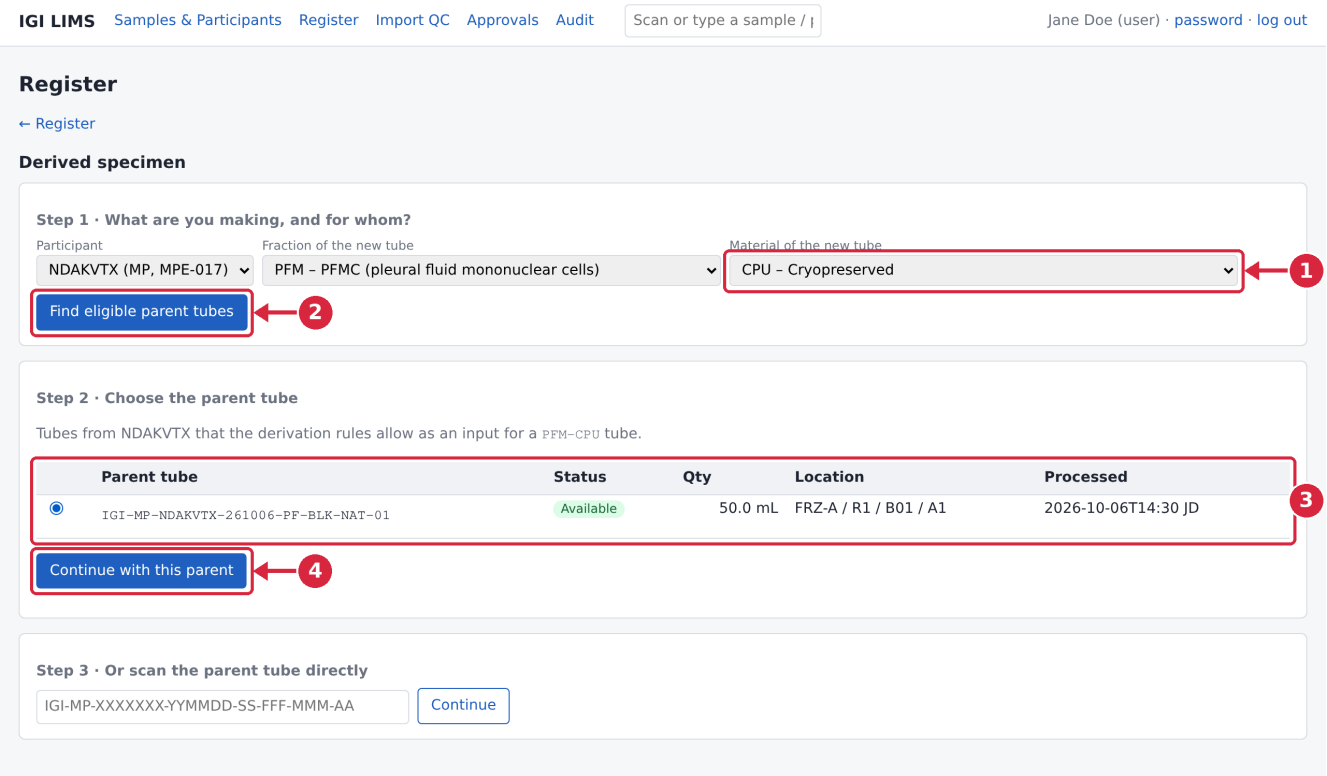
1. The issued ID: IGI-MP-participant-date-PF-BLK-NAT-01
2. Derive children directly from this tube
3. Metadata recorded on the form
4. Edit goes through two-person approval
5. Append-only hand-off log

6. Walkthrough: deriving tubes

A derived tube is any tube made from another: a cryovial from a cell preparation, DNA from a pellet, a library from DNA. The registration flow asks what is being made first and then offers only the parents that the rules allow, so the operator never has to work out eligibility by hand.

6.1 Choose what to make, then choose the parent

Registering a derived specimen starts from the product, not the parent. Here: PFMC (pleural fluid mononuclear cells), cryopreserved.

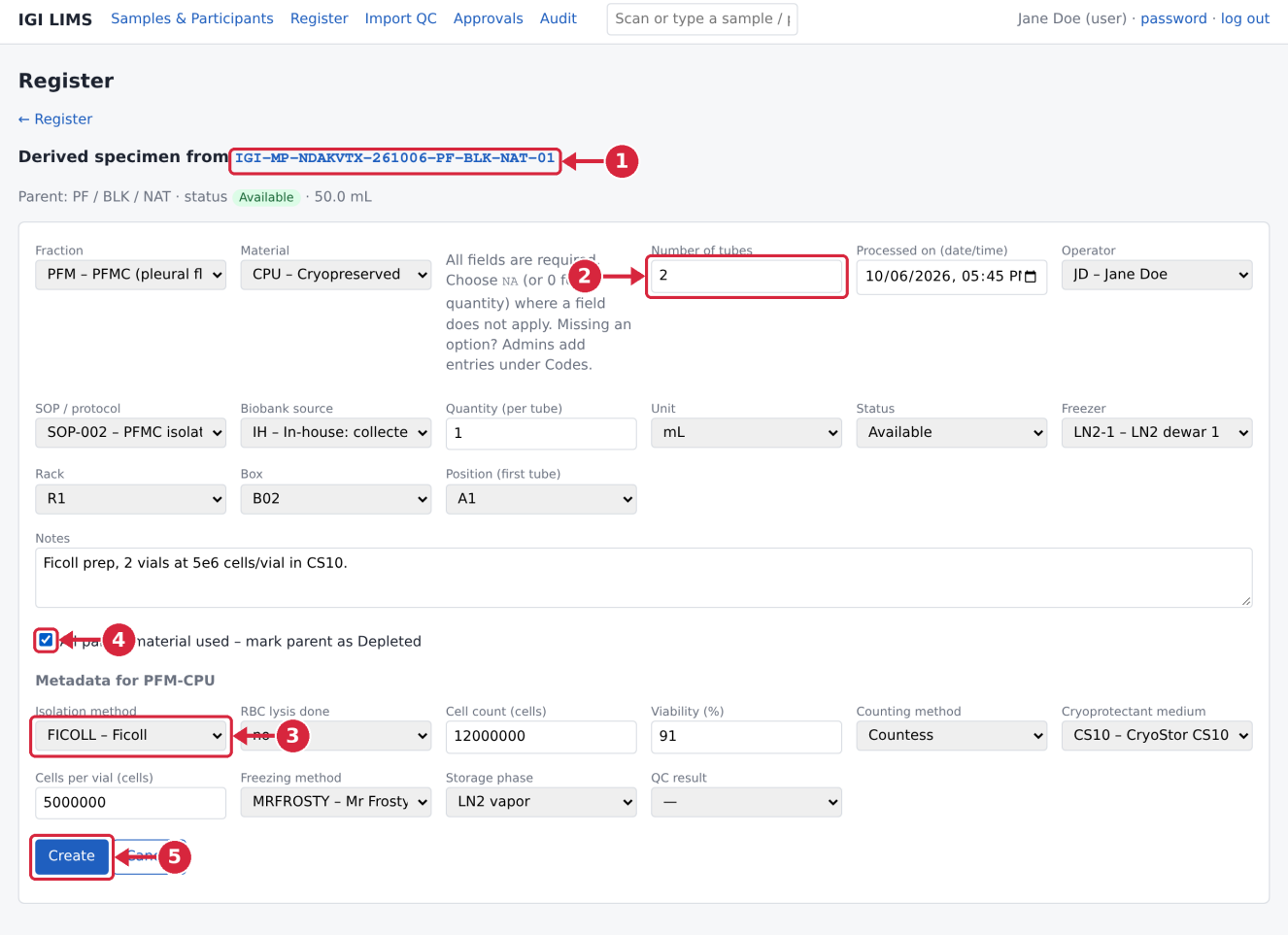


1. Say what the new tube will be: participant, fraction and material
2. Find eligible parent tubes
3. Only tubes the derivation rules accept as inputs are listed
4. Continue with the selected parent

6.2 Fill in the new tubes

Here two cryovials of PFMC are made from the pleural fluid. Metadata for a PFM fraction (isolation method, cell count, viability, counting method) and for a CPU material (cryoprotectant, cells per vial, freezing method, storage phase) appear together.

The derived-tube form. Fraction and material are fixed from step 1; the rest is filled in.

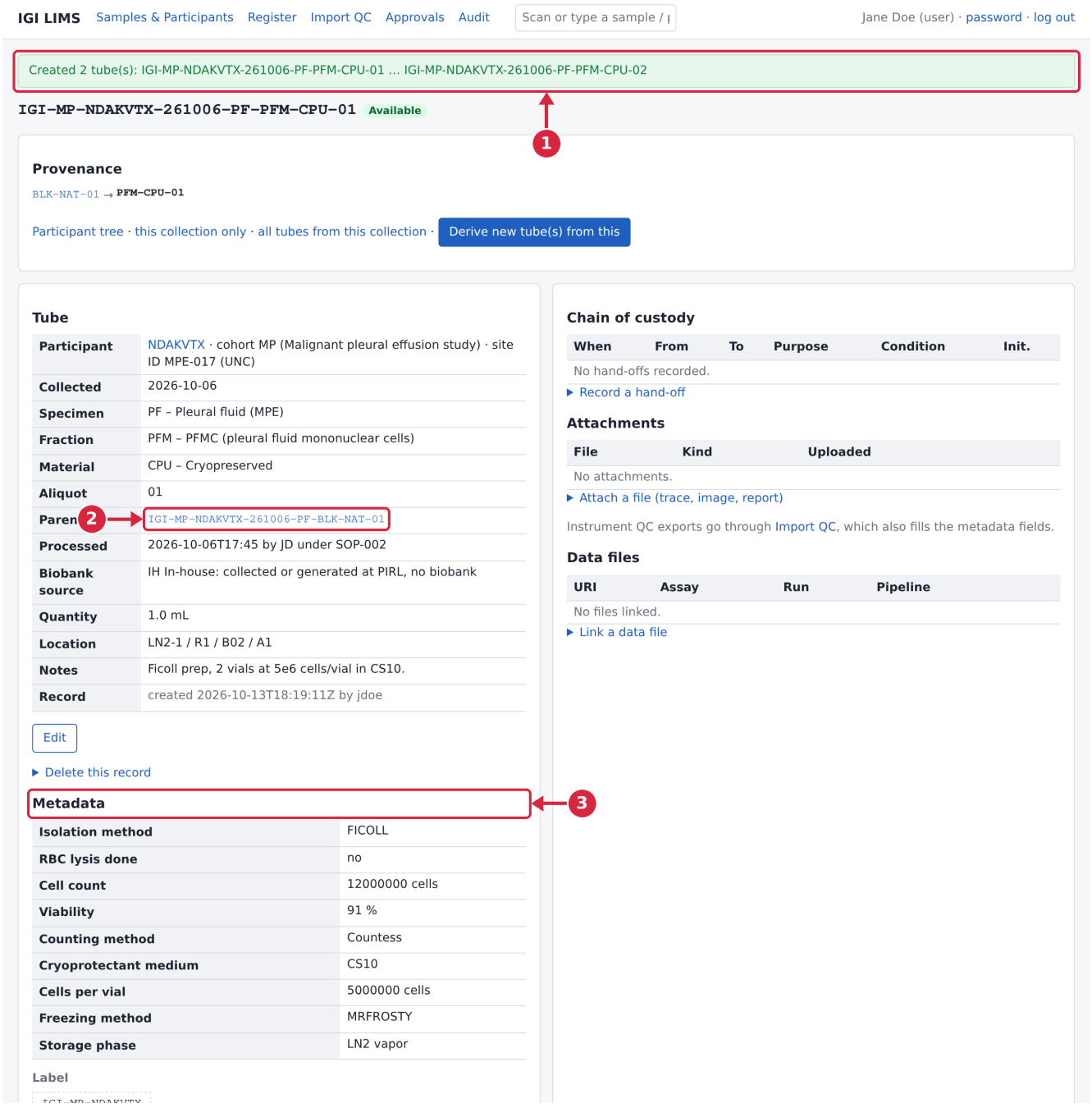


1. The parent tube
2. Number of tubes: two aliquots get consecutive numbers
3. Metadata for PFM and CPU: isolation, counts, viability, cryo medium
4. Marks the parent Depleted in the same step
5. Create

6.3 The result

The two tubes are IGI-MP-NDAKVTX-261006-PF-PFM-CPU-01 and IGI-MP-NDAKVTX-261006-PF-PFM-CPU-02. The parent, now marked Depleted, is one click away.

The first cryovial. Its parent is one click away, and so is every ancestor.



1 Two tubes created: ...-01 and ...-02

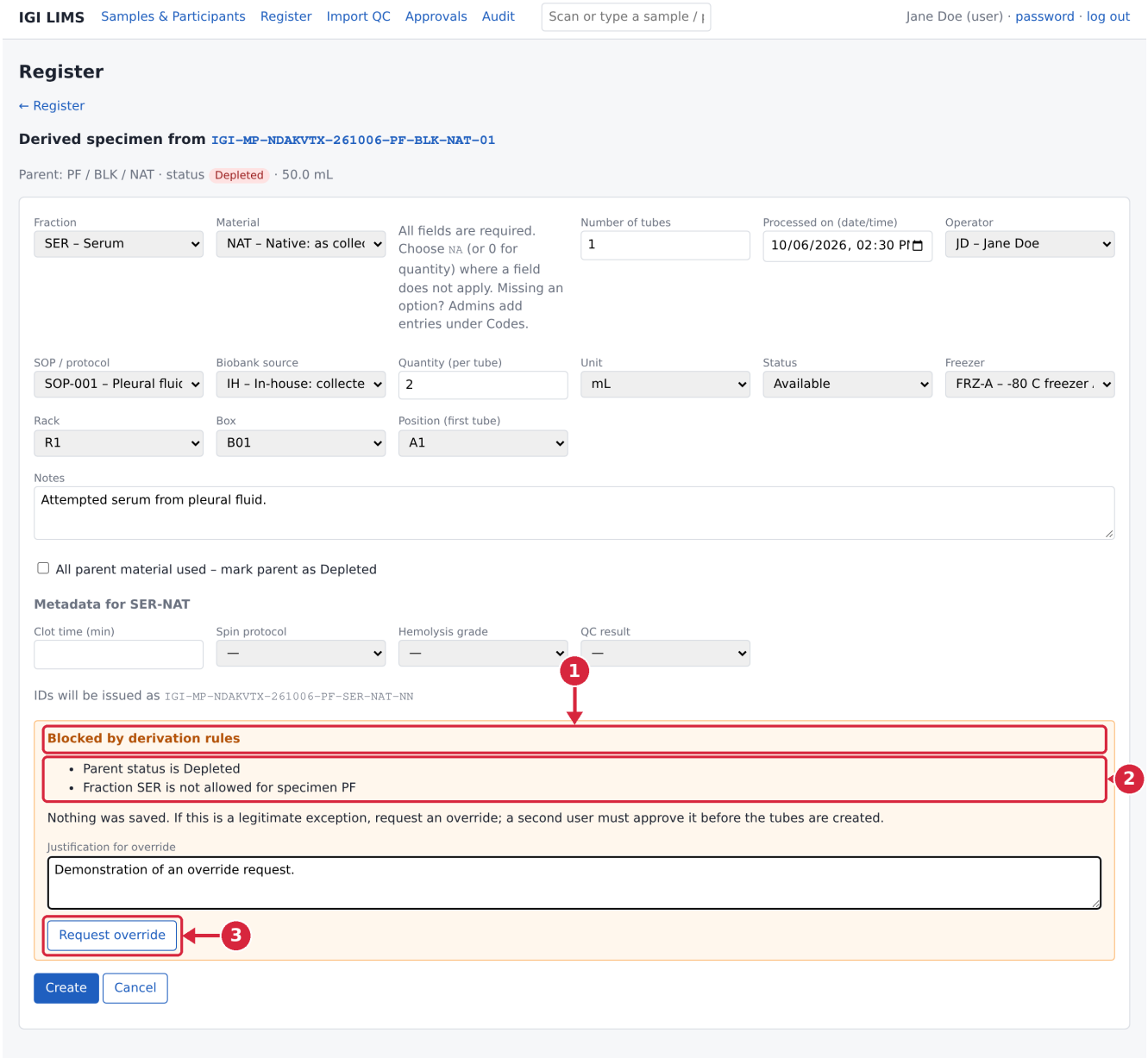
2 Parent link; the parent is now Depleted

3 Metadata stored with the tube

6.4 When the rules say no

Here the operator tried to register serum derived from pleural fluid, from a parent that is already depleted. The form refuses, lists the reasons, and offers an override request with a mandatory justification. A second user will see that request in the approval queue (section 8).

Trying to register serum from pleural fluid. The derivation rules refuse and offer an override request instead.



1 The rule table blocks this derivation; nothing was saved

2 The specific rules violated

3 Request override: sends it to a second user

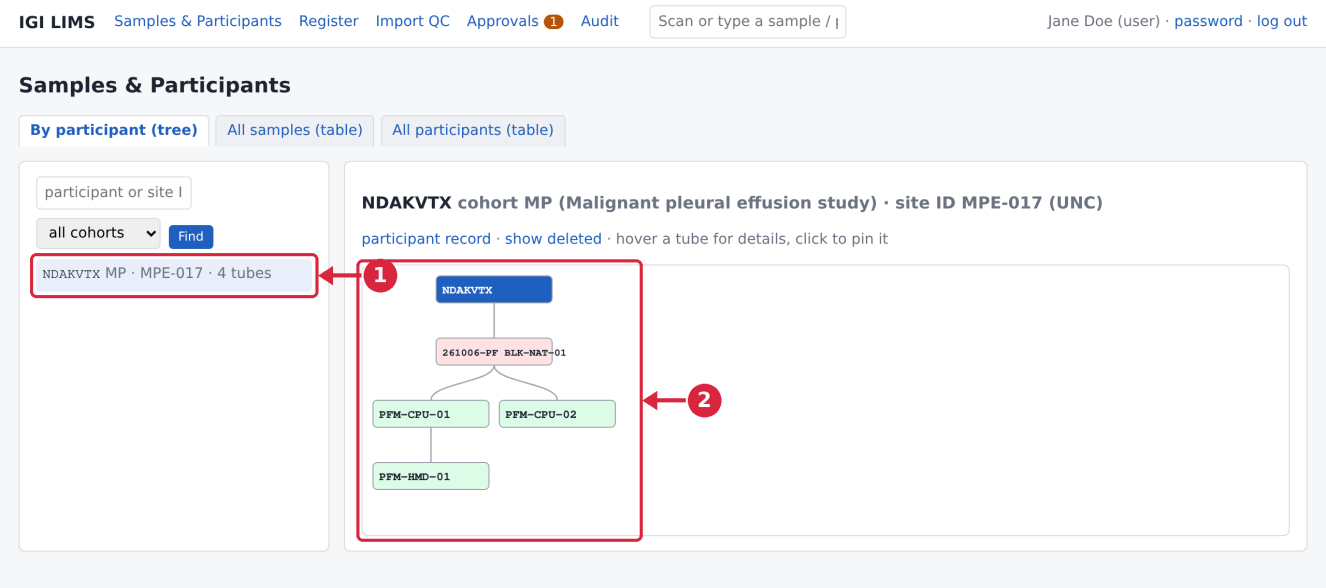
A third derivation, not shown as a form, extracted high-molecular-weight DNA from the first cryovial to make IGI-MP-NDAKVTX-261006-PF-PFM-HMD-01. It appears in the tree and is the tube edited in section 8.

7. Walkthrough: browsing

Samples & Participants is the one read-only page. It is safe to leave open on a shared screen, because nothing on it creates or changes a record.

7.1 The participant tree

Samples & Participants: a read-only view. Depleted tubes are shown with their status.

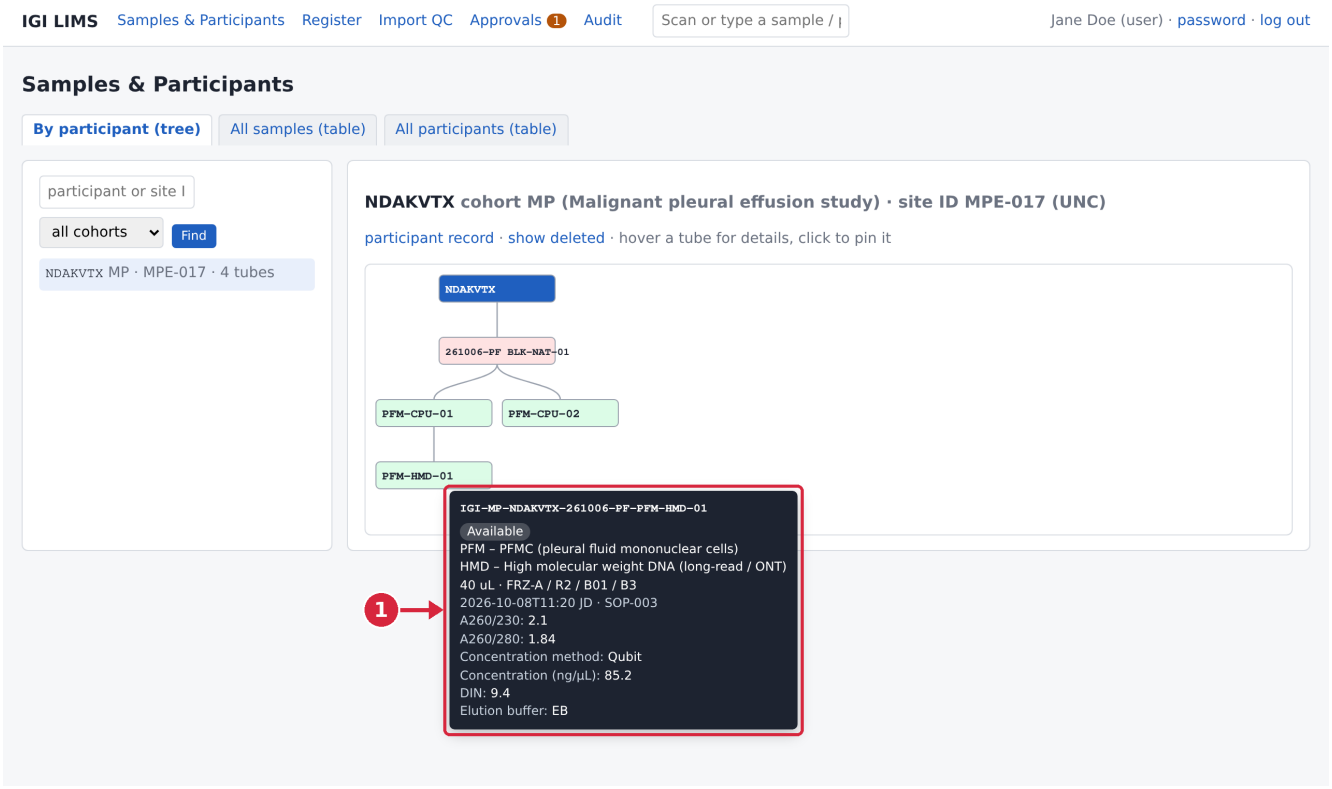


1 Pick a participant

2 Every collection and everything derived from it, top down

7.2 Hover for details

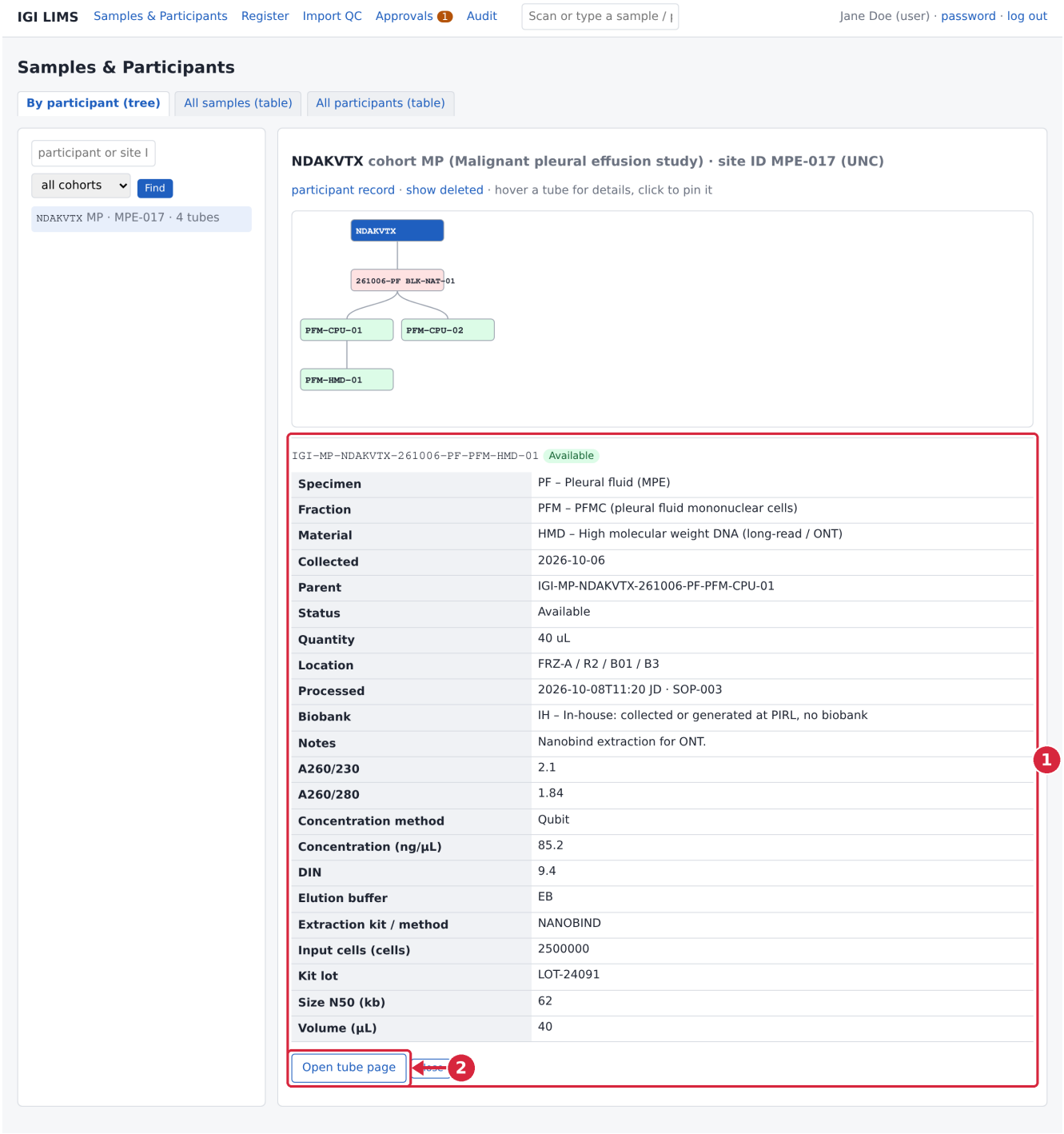
Hovering a node.



1 Hover shows status, quantity, location and key metadata

7.3 Click to pin

Clicking a node pins its details, with all metadata, and a link to the tube page.



1 Click pins the details

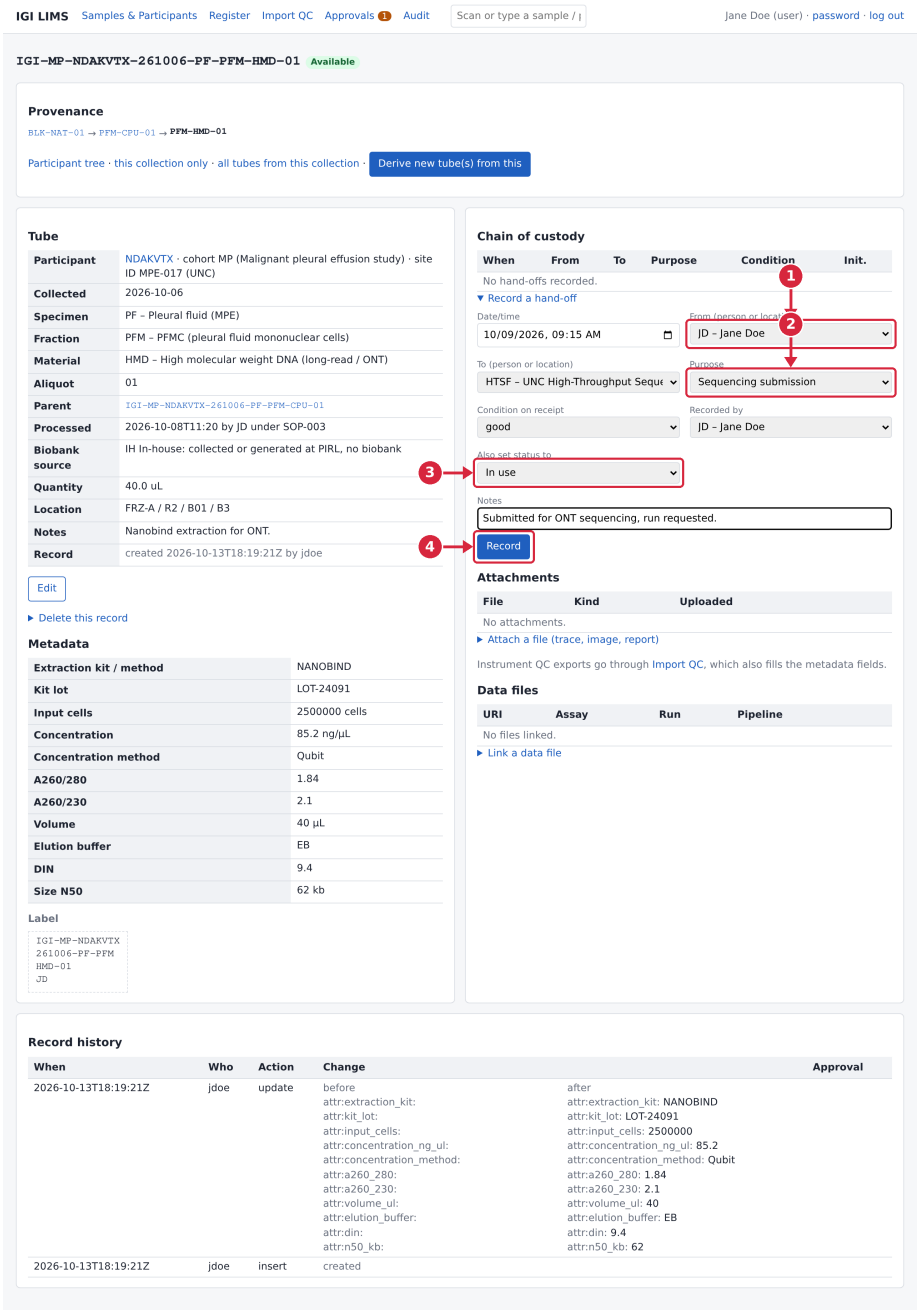
2 The only way out is to the tube page

8. Walkthrough: custody, editing and approval

8.1 Record a hand-off

Custody events are append-only. Each records from, to, purpose, time and condition. The DNA tube is handed to the sequencing core here and marked In use in the same step.

Recording a hand-off of the DNA tube to the sequencing core.



1 From: a user or a party such as HTSF or CCPD

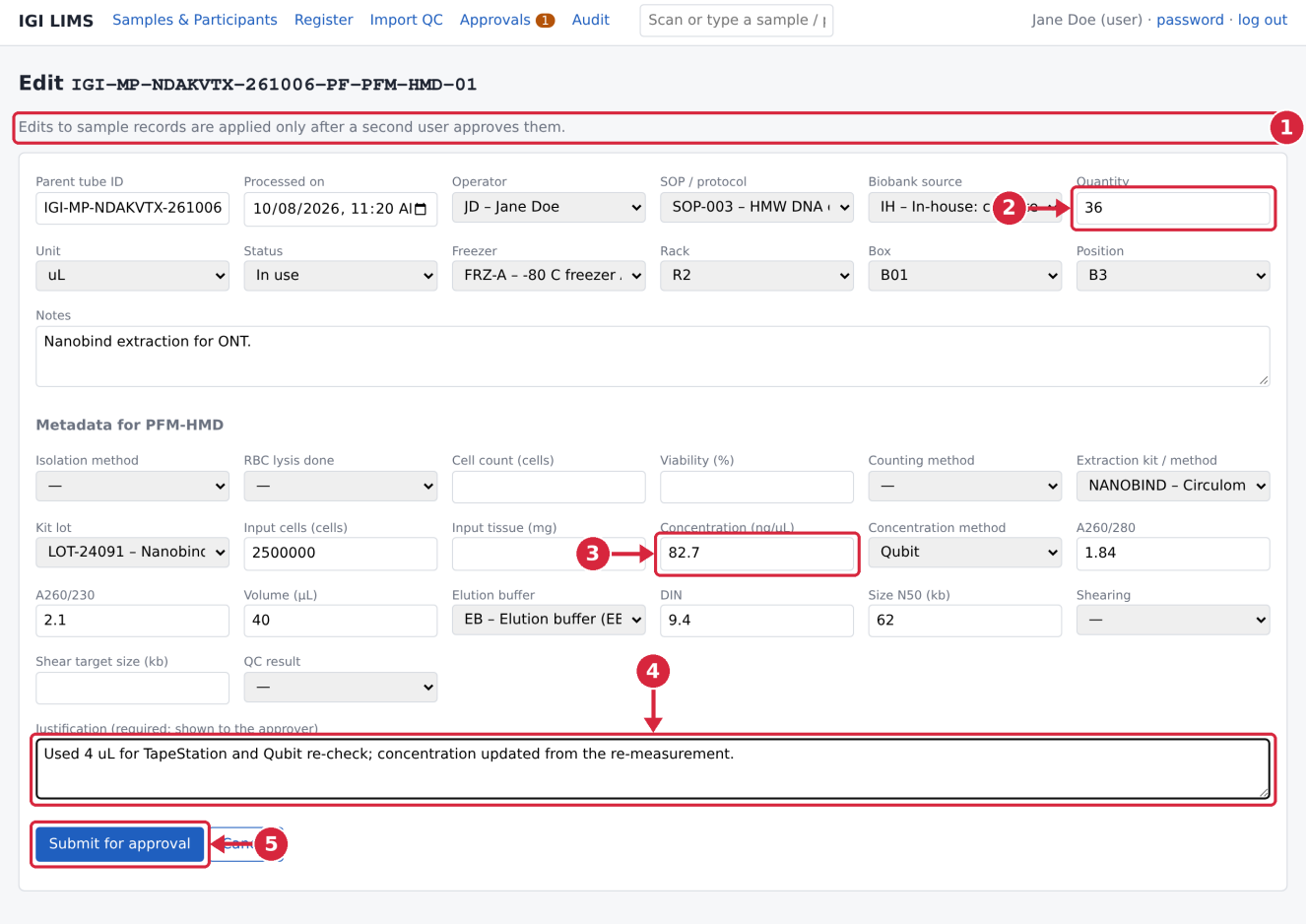
2 Purpose, from the custody purpose list

3 Optionally update the tube status in the same step

8.2 Modify a record

Jane Doe re-measured the DNA and used some of it for QC, so the quantity and the concentration need to change. The edit form is the same for tube fields and metadata.

Modifying a record. Nothing changes until someone else approves.



1 Edits are applied only after a second user approves

2 Quantity 40 to 36

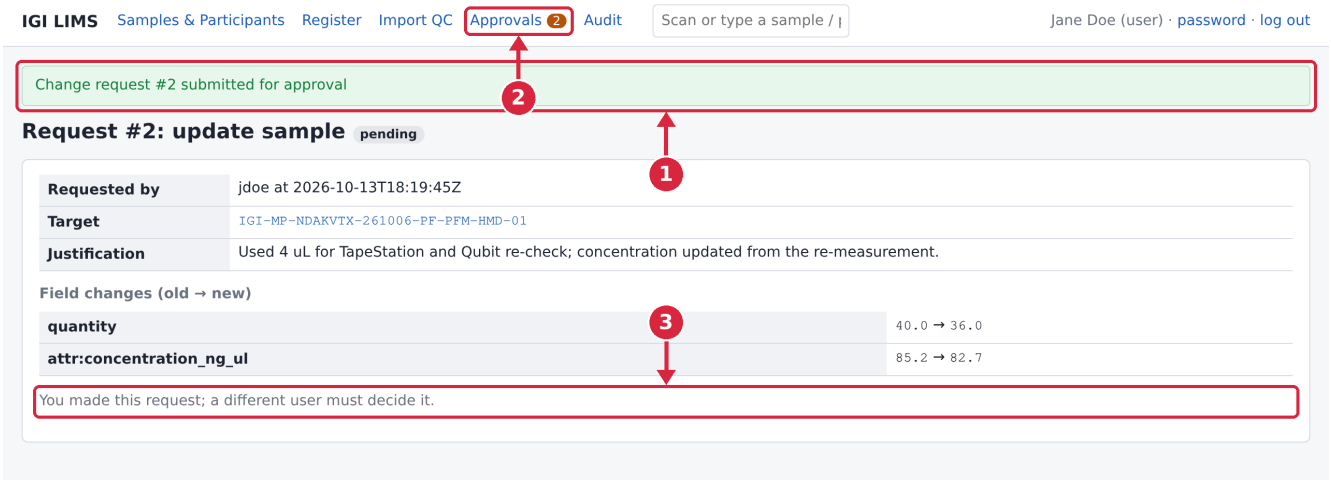
3 Metadata is edited on the same form

4 A justification is required and shown to the approver

5 Submit for approval

8.3 The request is queued

After submitting the edit, the requester lands on the request itself and can only wait.



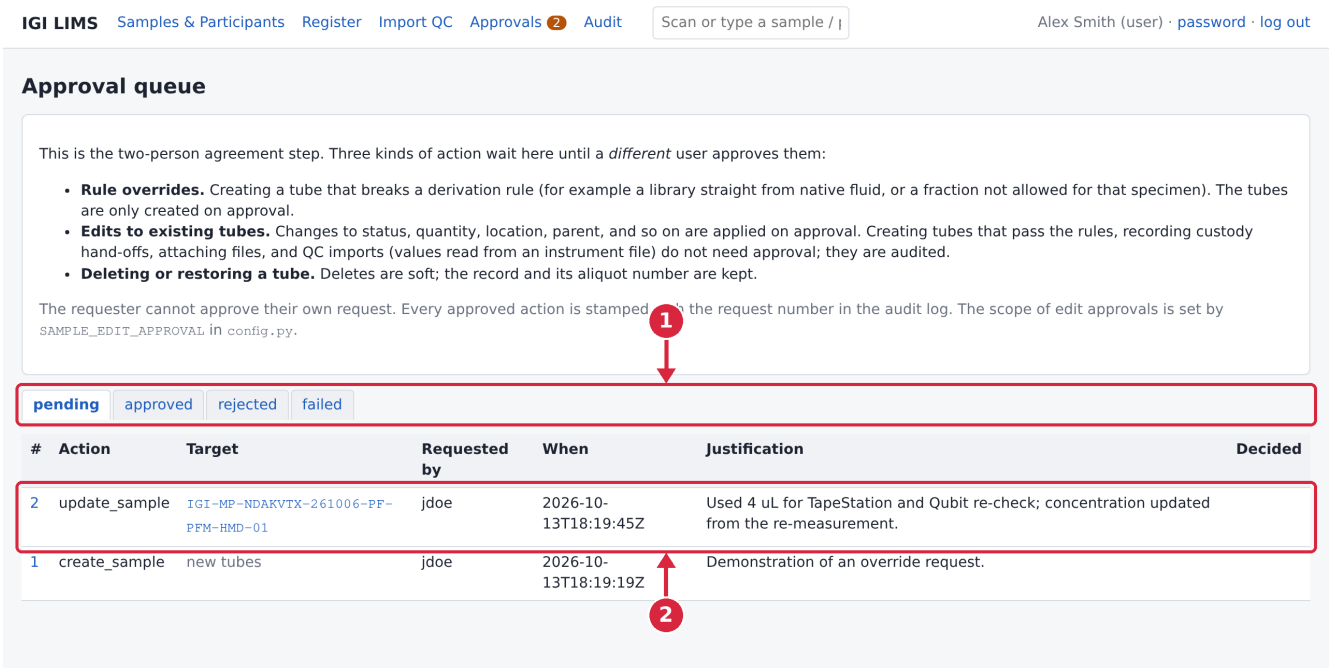
1 The request is queued; the tube still shows the old values

2 Pending count on the Approvals link

3 The requester cannot decide their own request

8.4 The approval queue, seen by a second user

Alex Smith's view of the queue. The requester cannot approve their own request.

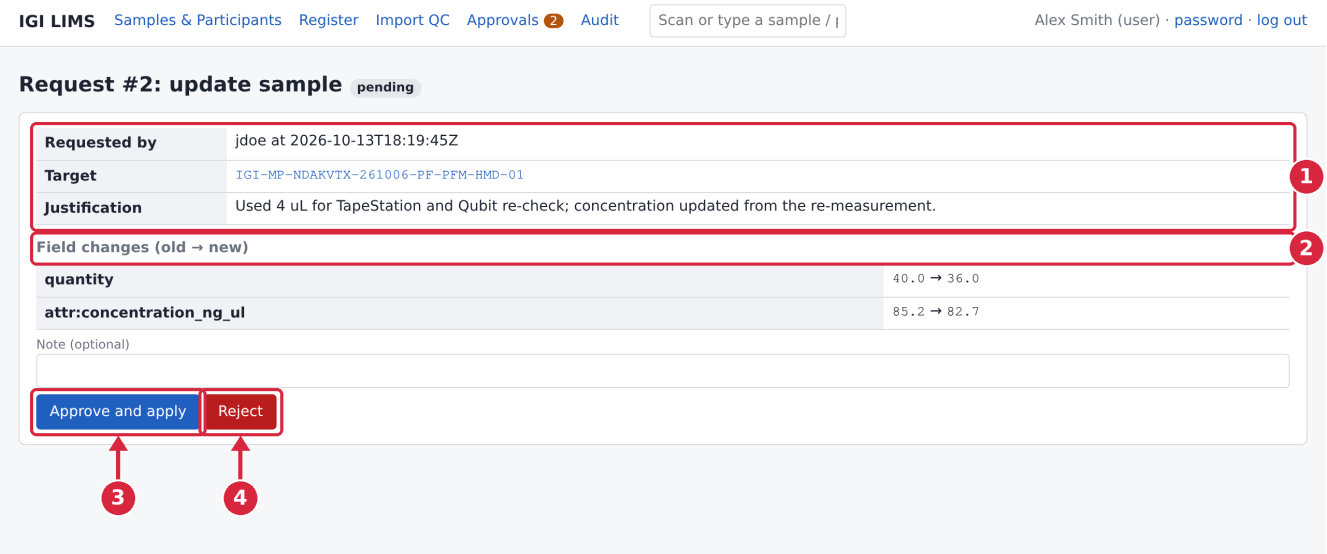


1 Pending, approved, rejected, failed

2 Each request: what, which tube, who asked and why

8.5 Reviewing a change request

The approver sees the exact diff before deciding.



1 Who, which tube, and the justification

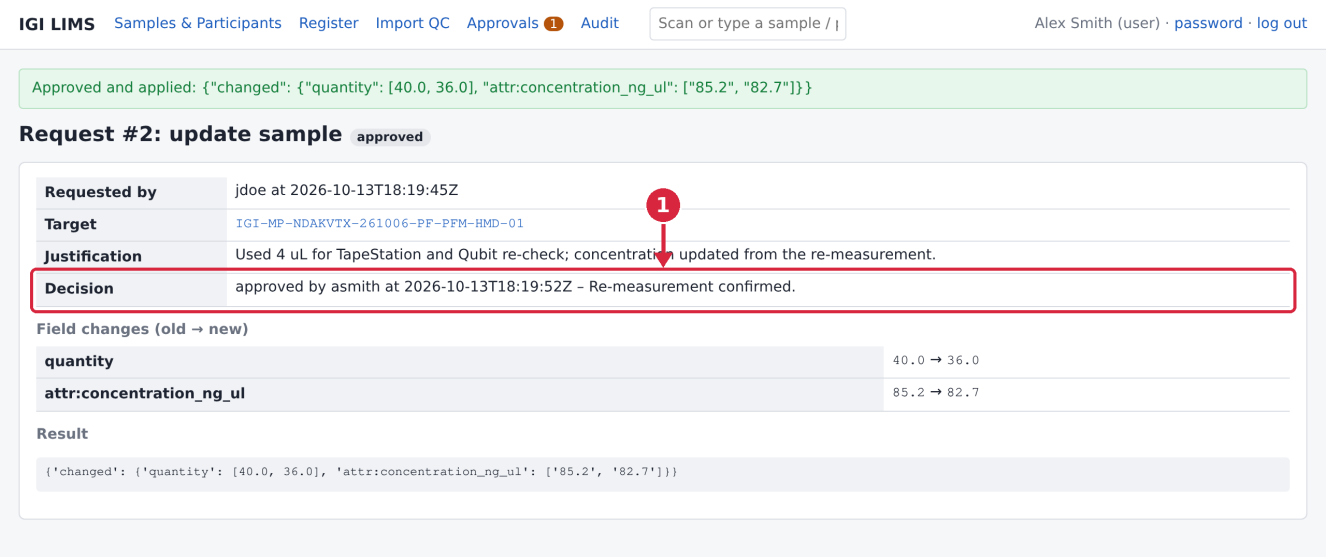
2 Exactly what will change, old to new

3 Approve and apply

4 Reject

8.6 After the decision

The decision is recorded on the request, and the change is applied to the tube.

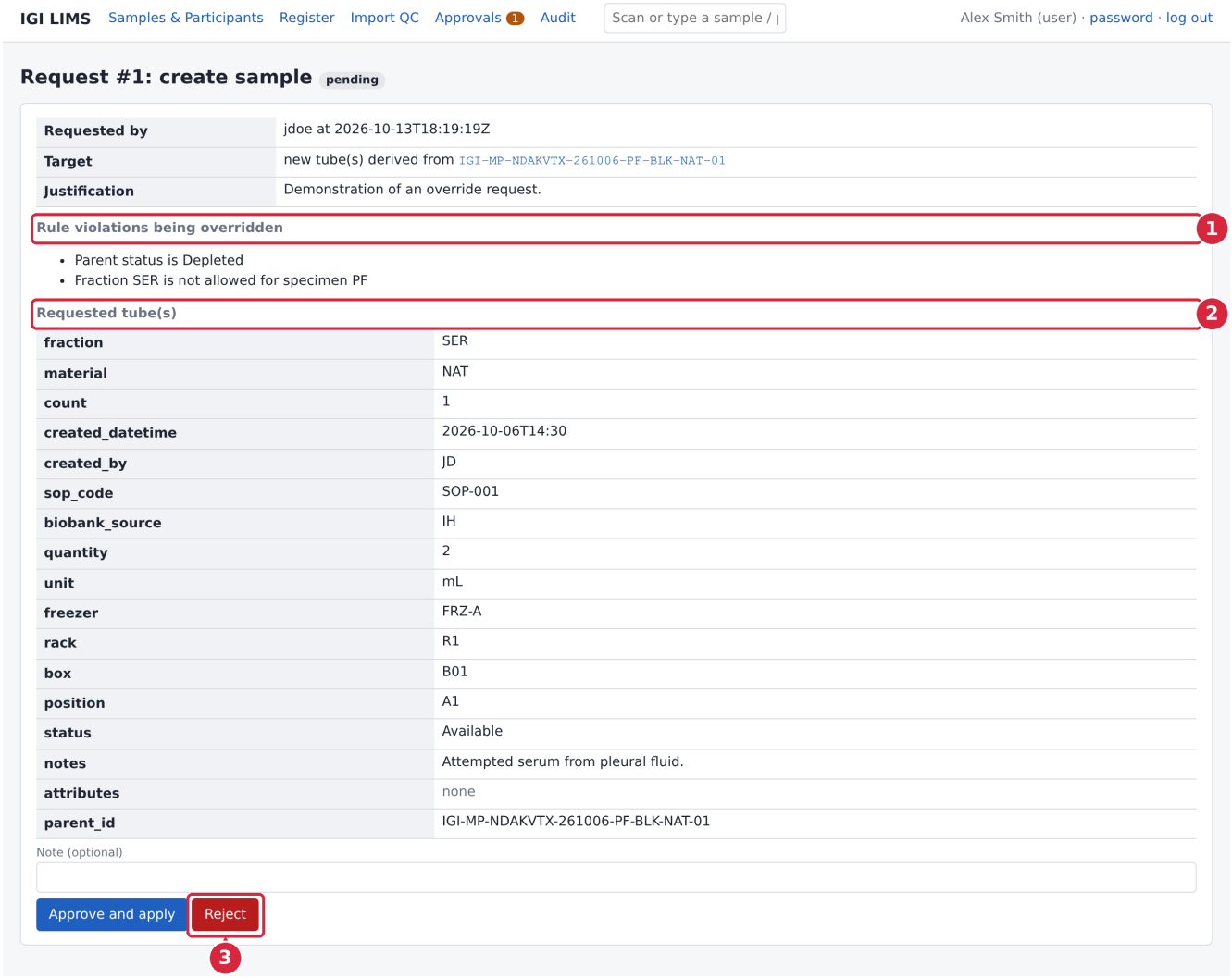


1 Decision, by whom, and when

8.7 Reviewing the override request from section 6.4

The same queue holds rule-override requests. The approver sees the rules that would be broken and the full payload of the tube that would be created. In this example the request was rejected with a note suggesting the correct fraction code.

The override request from the earlier rule block, as seen by the approver.



1 The rules that would be overridden

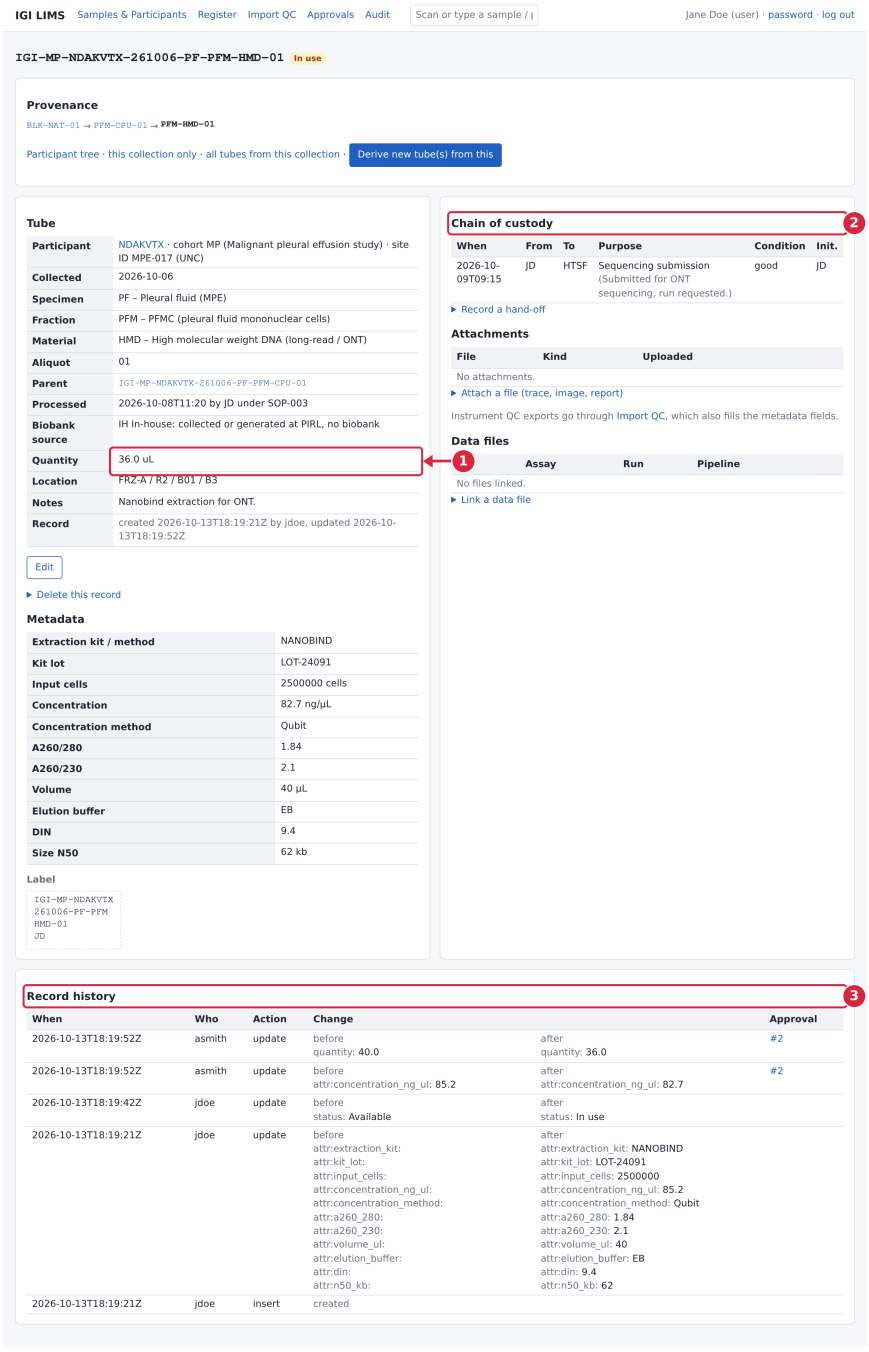
2 The full payload of the tube that would be created

3 Reject

8.8 The tube after the approved edit

Quantity and concentration now show the new values. The history lists the approved change with the approval number, the custody event's status change, and the original creation.

The DNA tube after the approved edit.



1 Quantity now 36

2 The hand-off recorded earlier

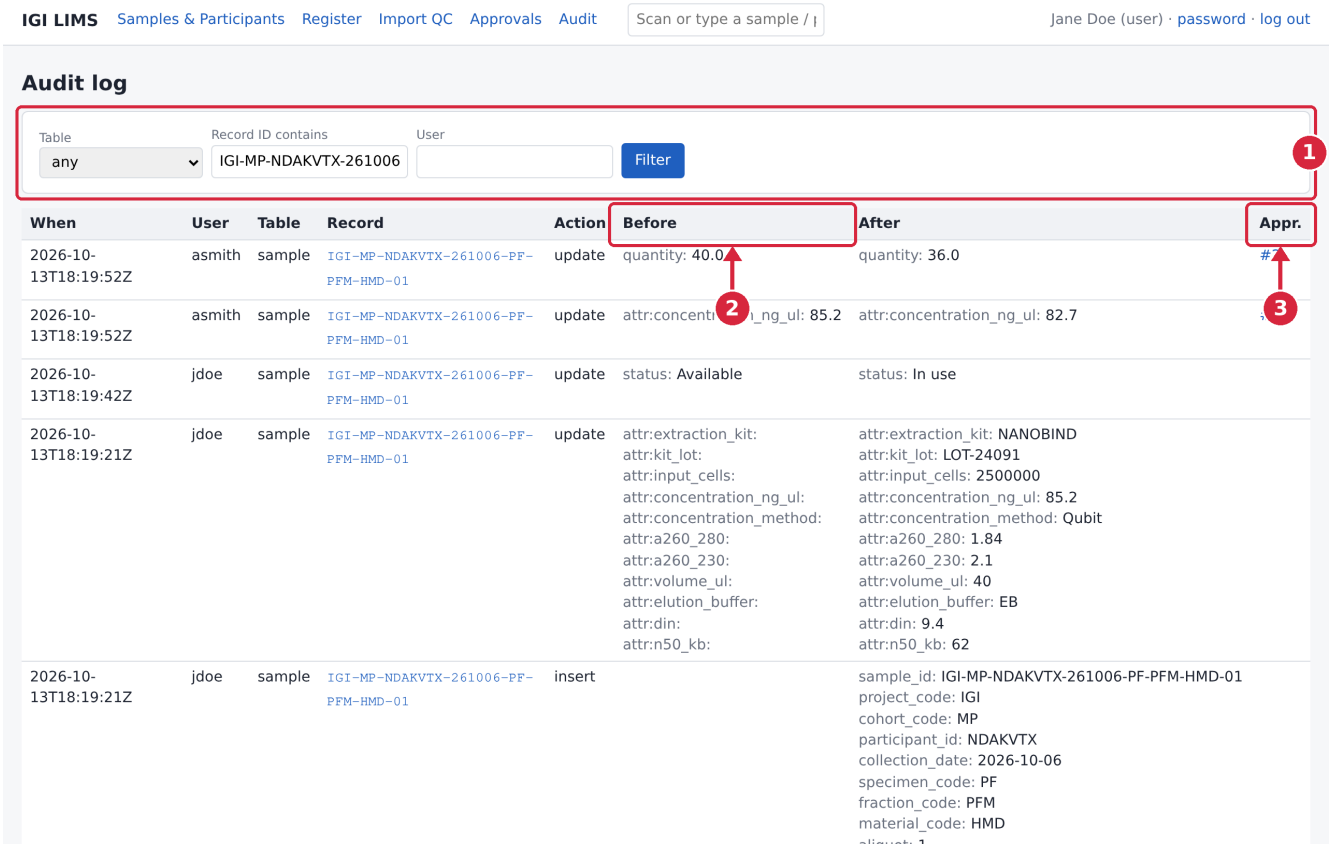
3 Every change with before and after values and the approval number

9. Walkthrough: the audit log

9.1 Everything that happened to one tube

The audit log is the system-wide version of the history block on each tube page. It covers participants, tubes, custody, data files, metadata, attachments, code lists, fields and users. Before and After are shown as one key and value per line so a reviewer can read them without parsing JSON.

The audit log filtered to one tube. Inserts, updates, custody and metadata writes all appear here.



1 Filter by table, record ID or user

2 Before and After as key: value lines

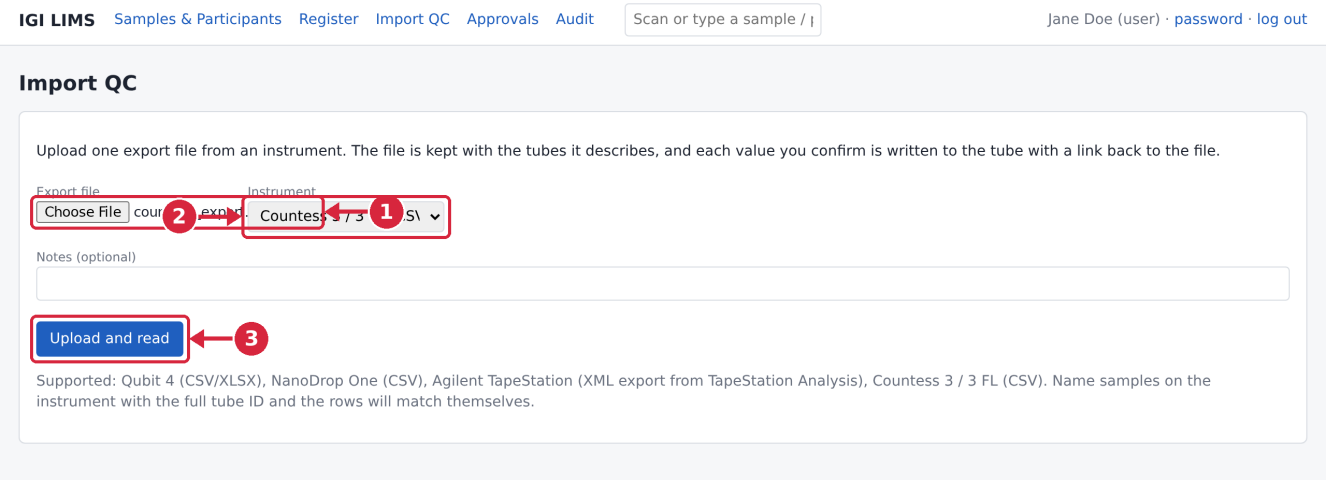
3 Link to the approval that authorised the change

10. Walkthrough: importing instrument QC files

Common instrument exports can be loaded directly instead of typing numbers into metadata fields. Supported today: Qubit 4 and NanoDrop One exports (parsed through the allotropy library), TapeStation XML, and Countess 3 CSV. The example below imports a Countess cell count.

10.1 Upload one export

Importing a Countess 3 cell-count export.



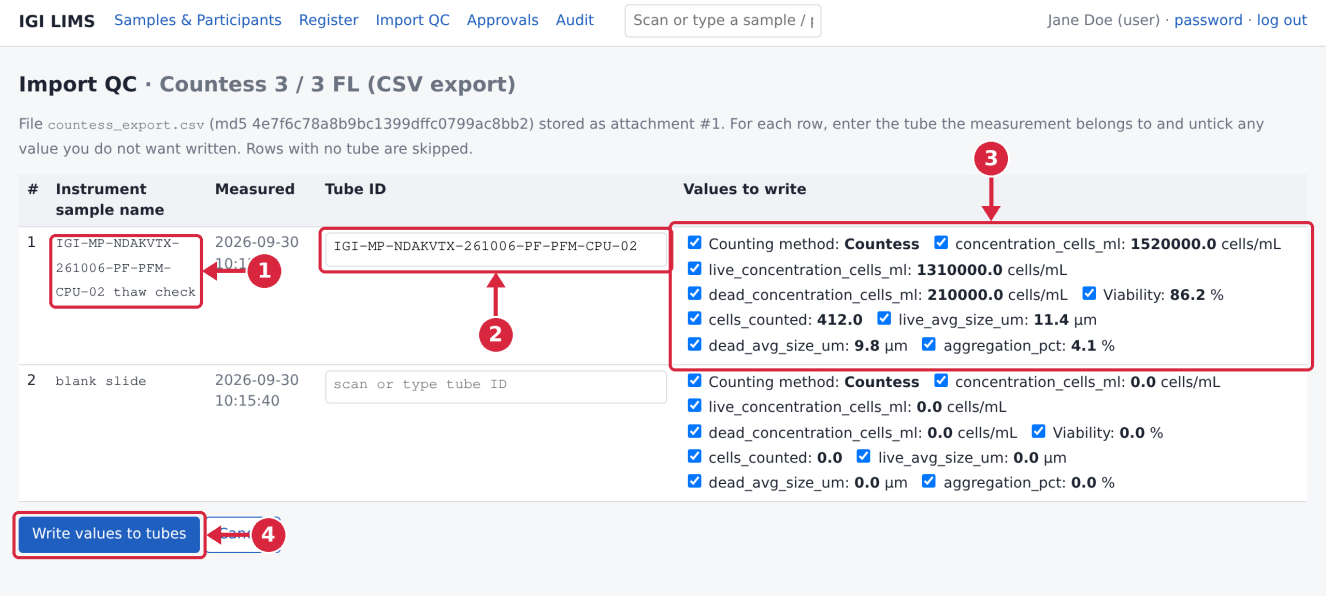
1 One instrument export file

2 Instrument: auto-detected if left blank

3 Upload and parse

10.2 Confirm what goes where

One row per instrument sample. Rows with no tube ID are skipped; the blank slide stays blank.



1 The instrument's sample name

2 Suggested tube ID: matched because the name contains it

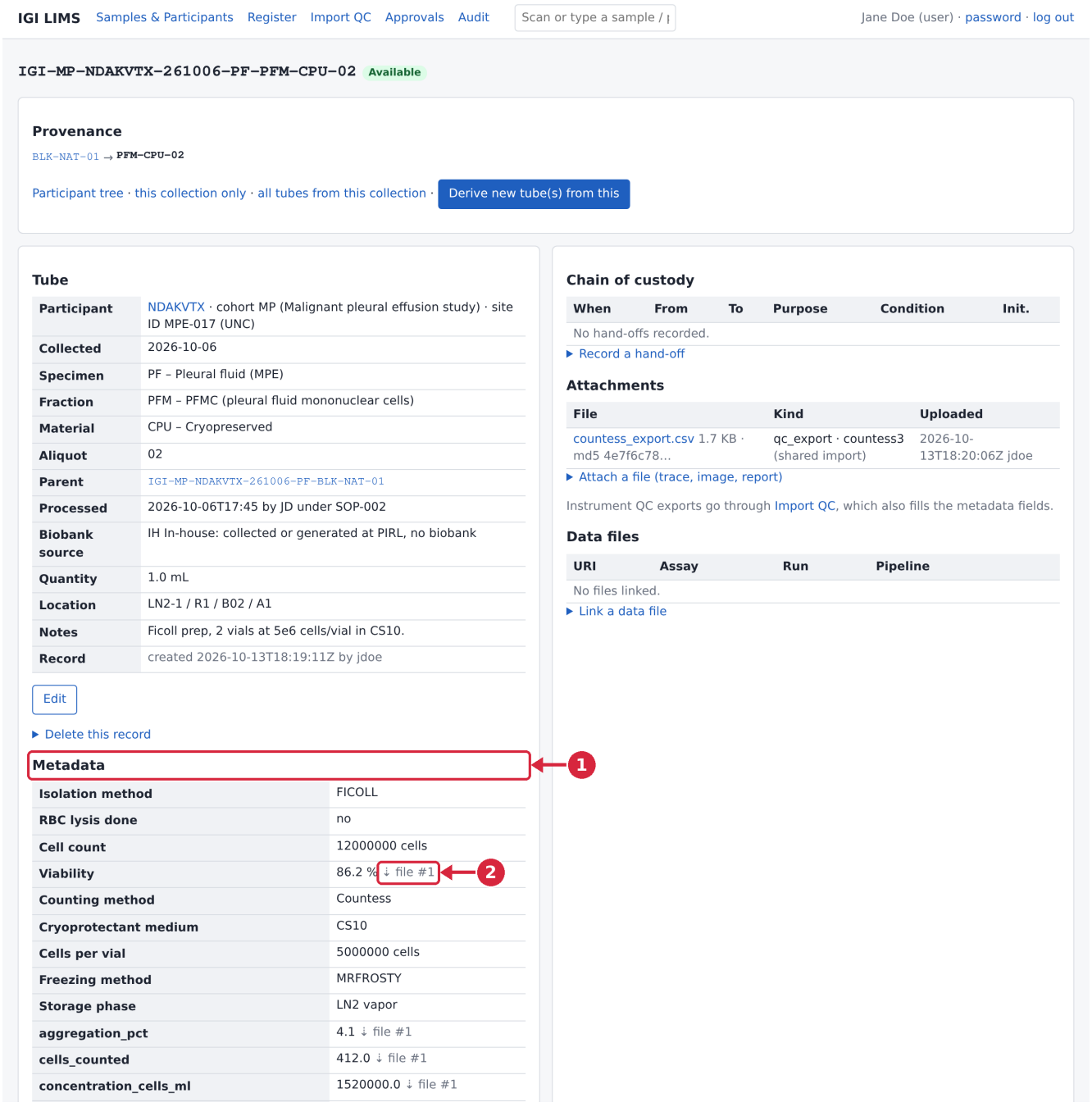
3 Untick any value you do not want written

4 Write values to tubes

10.3 Provenance of imported values

Each value written by an import keeps a link to the stored export file, so a concentration or viability can always be traced to the instrument run it came from. The export itself is kept as an attachment.

The second cryovial after import. Each imported value points back to the file it came from.



1 Imported values, with a link to the stored export file

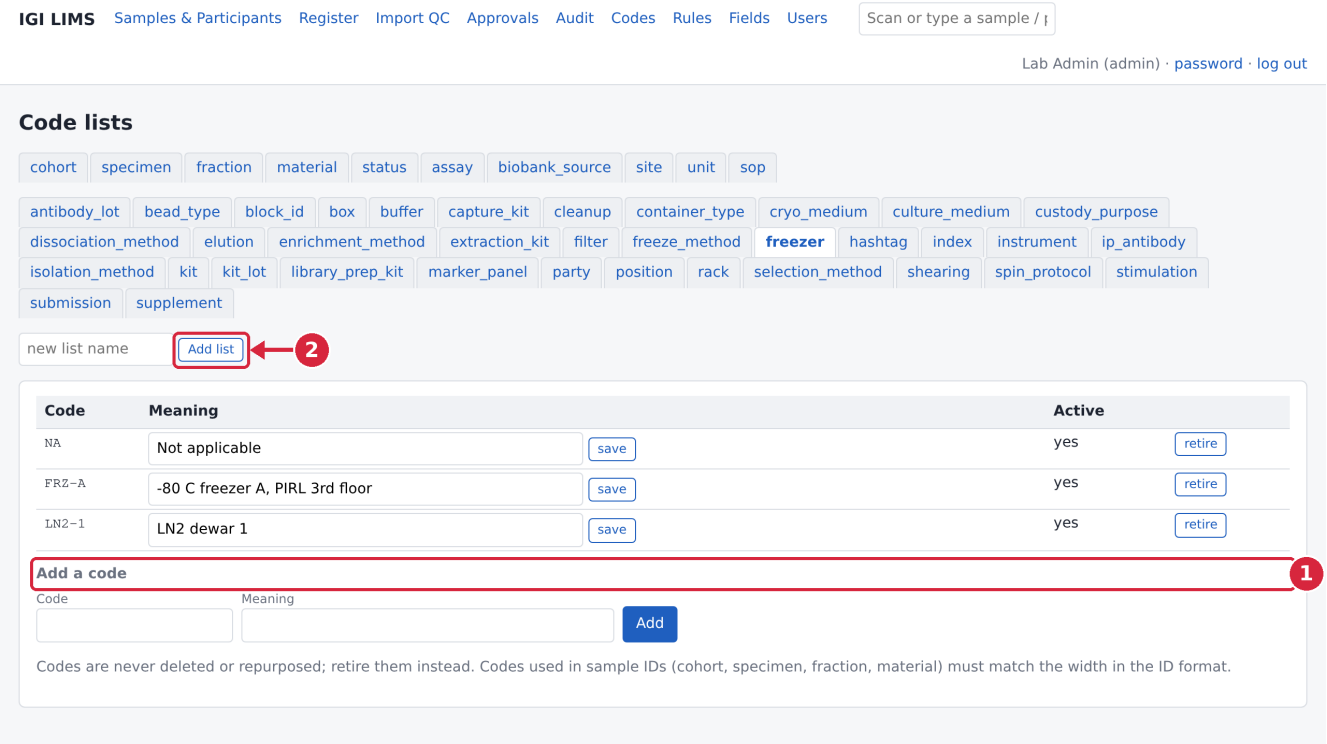
2 Source file

11. Administration

Administrators see four extra pages. None of them require a code change or a restart.

11.1 Codes: every controlled vocabulary

The Codes page. Entries are retired, never deleted, so old records keep their meaning.



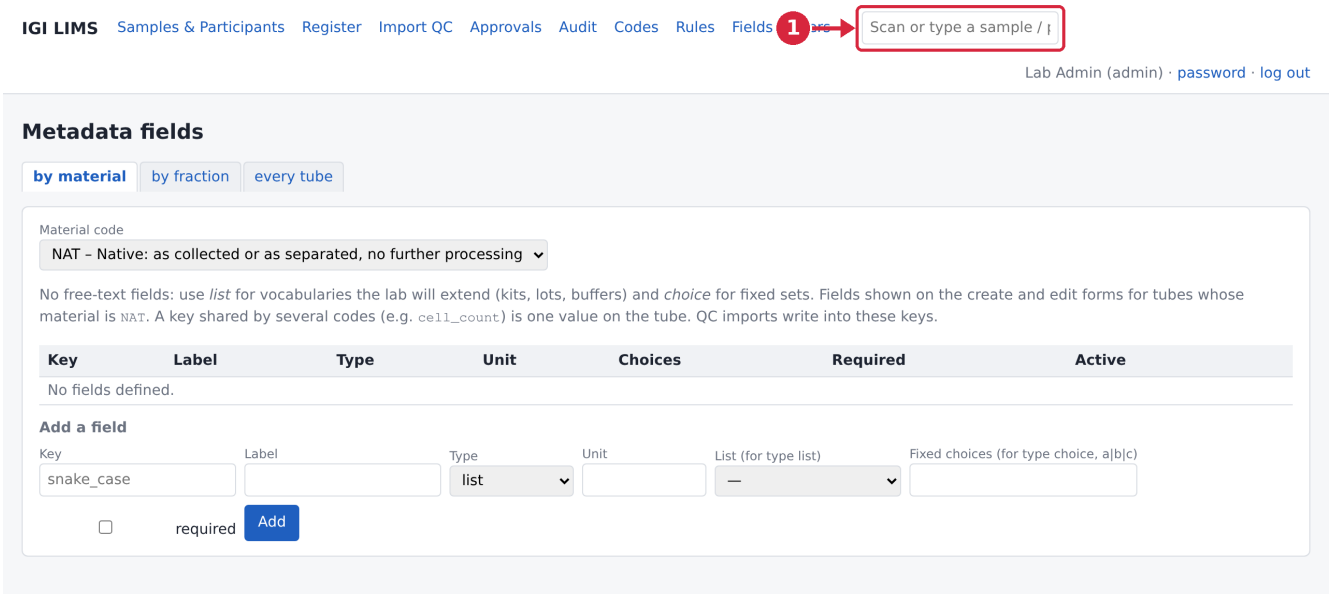
1 Add an entry to any list

2 Create a whole new list for a new metadata field

11.2 Fields: which metadata each fraction and material carries

Fields are typed as number (with unit), fixed choice, admin-extendable list, yes/no, date or date-time. There is no text type. A field can be marked required.

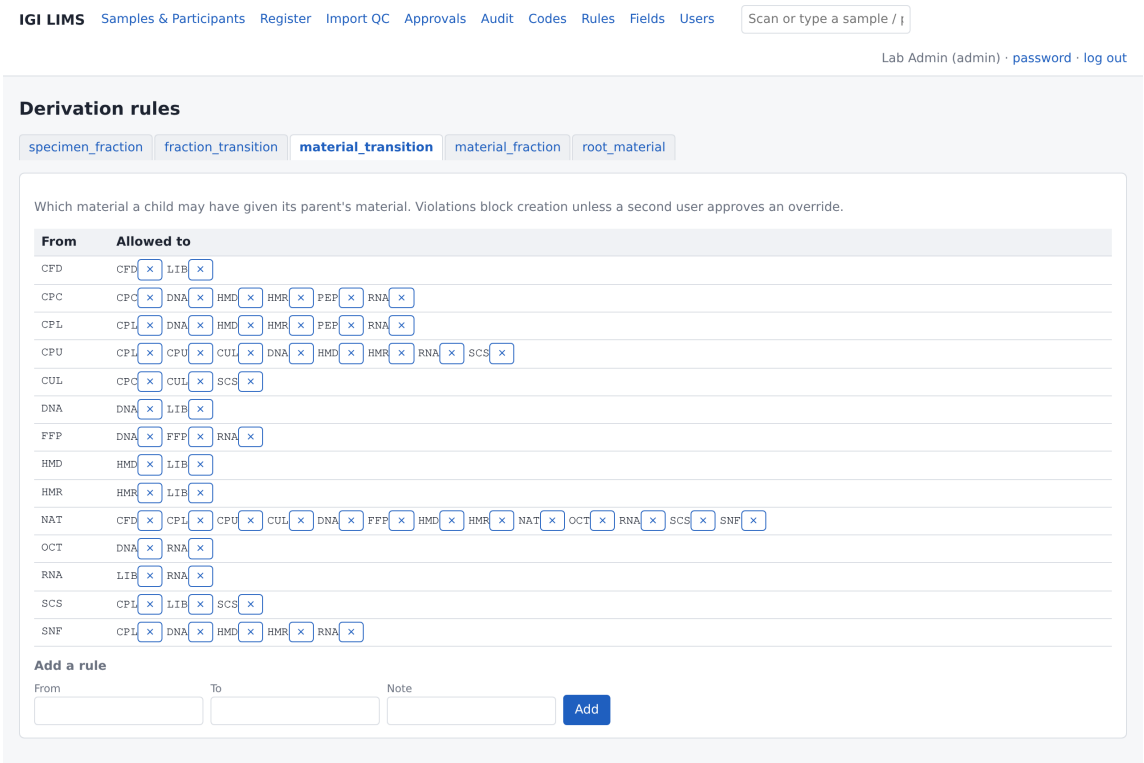
The Fields page defines which metadata each fraction or material carries.



1 Add a metadata field: scope, label, type, unit, list, required

11.3 Rules: allowed specimens, fractions and transitions

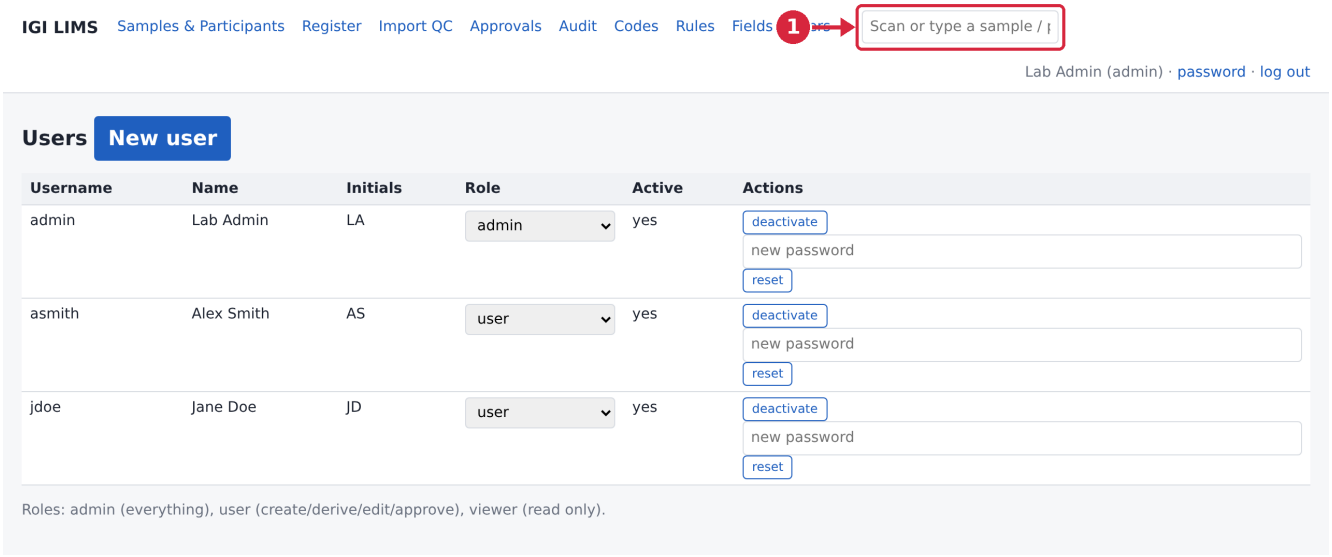
The Rules page: which fractions a specimen may have, and which transitions are allowed.



11.4 Users and roles

Roles are admin (everything), user (create, derive, edit, approve) and viewer (read only).

The Users page.



1 Create a user with a role: admin, user or viewer

12. Also available

Data files

Sequencing and mass-spectrometry outputs live in Google Cloud Storage buckets. A data file record links a bucket URI to the tube that was sequenced or run (usually a library or a peptide eluate), with assay, run ID, checksum and pipeline version, so a file is always reachable from the collected specimen it came from. Files can be linked from the tube page or by a pipeline through the API.

Attachments

Any file (trace, image, report) can be attached to a tube. It is stored on disk under the attachment directory with an md5 checksum and listed on the tube page.

Read-only JSON API

For pipelines and scripts. Authenticate with a browser session or an `X-Api-Token` header matching the token set in the server environment.

Endpoint	Returns
/api/pattern	anchored regex and fixed length for IDs
/api/validate/<id>	the ID parsed into components, and whether it exists
/api/samples?participant=&root=&material=&status=&prefix=	list of tubes
/api/samples/<id>	one tube with custody events and files
/api/samples/<id>/lineage	ancestors (root first) and all descendants
/api/tree/<root_id>	nested tree for one collection
/api/files	data files with their tube and lineage

Technical summary

- Python 3.10, Flask 3, Jinja templates, a small amount of vanilla JavaScript for the tree and dynamic forms.
- SQLite in WAL mode with foreign keys on; one file database, backed up by copying the file.
- Tables: code_list, derivation_rule, user, participant, sample, custody_event, data_file, approval_request, audit_log, attribute_definition, sample_attribute, attachment.
- Instrument parsing via allotropy (Benchling open source) for Qubit and NanoDrop; TapeStation and Countess parsed directly.
- 38 automated tests cover ID composition, the registration and approval workflow, parsers, import, metadata and vocabulary enforcement.
- Deployment target: a GCP VM behind gunicorn, reachable by collaborators over the web; not yet set up.

13. Open questions (tracked in NOTES.md)

- Participant code format: seven letters as specified, or a site-style code such as P101.
- NTC and NRM are three-character specimen codes but the ID segment is two characters.
- Whether every edit should require two-person approval, or only provenance changes.

- The seeded derivation rules are a proposal and need review by the lab.
- Which metadata fields should be required.
- Most admin-extendable lists start with only NA and need real freezers, racks, boxes, SOPs and kits before first use.
- The Countess parser follows the vendor column specification and has not yet been checked against a real export.